AI-Powered Agriculture Crop Advisory Assistant

Dataset Details Document

1. Dataset Overview

The system uses a synthetic agricultural dataset of 1,500 records generated to mirror the distribution of real Indian agro-climatic data from ICAR, IMD, and State Agriculture Departments. Each record represents a historical crop trial case.

2. Dataset Statistics

Attribute	Value
Total Records	1,500 cases
Feature Dimensions	13 input features + 1 target
Crops Covered	12 major varieties (Rice, Wheat, Cotton, Maize, Sugarcane, Chickpea, Groundnut, Soybe
States Covered	22 Indian states
Seasons	Kharif, Rabi, Whole Year, Summer, Autumn, Winter
Year Range	Synthetic data aligned to 2015–2024 agro-climatic patterns
Missing Values	None — all records complete
Train / Test Split	80% / 20% (1,200 train, 300 test)

3. Feature Schema

Feature	Type	Unit	Range	Description
Crop	Categorical	—	12 classes	Target crop variety
Season	Categorical	—	6 seasons	Agricultural season
State	Categorical	—	22 states	Indian state of cultivation
Area	Continuous	hectares	0.1 – 500	Cultivated land area
Nitrogen (N)	Continuous	kg/ha	0 – 500	Soil nitrogen content
Phosphorus(P)	Continuous	kg/ha	0 – 400	Soil phosphorus content
Potassium (K)	Continuous	kg/ha	0 – 400	Soil potassium content
pH	Continuous	—	4.0 – 10.0	Soil acidity/alkalinity
Rainfall	Continuous	mm	50 – 2500	Annual precipitation
Temperature	Continuous	°C	8 – 48	Mean growing temperature

Humidity	Continuous	%	20 – 95	Relative humidity
Fertilizer	Continuous	kg/ha	0 – 800	Total fertilizer applied
Pesticide	Continuous	kg/ha	0 – 50	Pesticide usage
Yield	Continuous	t/ha	0.5 – 120	TARGET: Recorded crop yield

4. Data Generation & Preprocessing

- Synthetic generation using scripts/generate_sample_data.py with seeded NumPy distributions
- Crop-specific agro-climatic parameter ranges drawn from ICAR crop production guides
- State-season-crop yield ranges calibrated against MOSPI agricultural statistics
- Text-based case IDs assigned (CASE_0001 to CASE_1500)
- No PII or personally identifiable farming data included

5. Storage & Indexing

- Raw CSV: data/raw/agricultural_data.csv
- Processed Parquet: data/processed/agricultural_data.parquet
- FAISS binary index: indexes/faiss_index.bin
- TF-IDF keyword index: indexes/tfidf_index.pkl
- Feedback store: data/feedback_store.json